Count My Fish: A Data-Driven Recreational Fishing Application

Prototype Recreational Fisheries Data Collection Application

Presented to:
North Carolina Division of Marine Fisheries

Presented by: Rebecca Rickard
East Carolina University – M.S. Data Science

Overview

Count My Fish is a data-driven recreational fishing application designed to improve both the angler experience and fisheries data collection through real-time reporting, predictive analytics, and community engagement. Developed as a prototype using Python and Streamlit, the app allows anglers to log catches, track fishing conditions, and receive fishing success predictions while contributing valuable environmental and behavioral data.

Traditional fisheries reporting methods are often delayed and limited in scope. Count My Fish addresses this issue by collecting real-time recreational fishing data directly from users. The application combines catch logging, environmental conditions, and predictive analytics into a single platform that benefits both anglers and fisheries management agencies.

Purpose and Objectives

The purpose of Count My Fish is to create a smarter and more connected fishing ecosystem where recreational anglers and fisheries agencies both benefit from shared data collection and analysis.

For anglers, the app provides tools to improve fishing success, including fishing predictions, interactive maps, catch tracking, and community competitions. Users can monitor their catches over time and participate in challenges and leaderboards that encourage continued engagement.

For fisheries management agencies, the app provides a framework for collecting real-time recreational fishing data that could support:

- Regional fishing trend analysis
- Species monitoring
- Seasonal behavioral patterns
- Environmental condition tracking
- Predictive fisheries analytics

This information could help agencies better understand recreational fishing activity and support future conservation and management efforts.

Core Features and Functionality

Catch Logging System

The app allows users to record detailed information about fishing trips and catches, including:

- Fish species
- Fish length
- Catch location
- Time of day
- Bait and fishing technique
- Weather and tide conditions

These records create a growing dataset that can later be analyzed for trends and predictive modeling.

Predictive Fishing Report

One of the main features of Count My Fish is the Fishing Report system, which predicts the likelihood of fishing success based on environmental and user-provided data. Various machine learning models were explored during development, including:

- Logistic Regression
- Random Forest
- XGBoost

The goal of the Fishing Report is to help anglers make more informed decisions about when and where to fish.

Interactive Mapping and User Engagement

The app includes an interactive map that displays fishing activity and catch locations. Users can filter catches by species and explore regional fishing patterns visually. To encourage long-term participation, the app also includes:

- Challenges and achievements
- Leaderboards
- Competition tracking

These features increase user engagement while encouraging consistent data reporting.

Value for Fisheries Management

Count My Fish has the potential to support fisheries agencies by providing near real-time recreational fishing data. If expanded into a production-level system, the application could support:

- Recreational harvest monitoring
- Species distribution analysis
- Environmental impact studies
- Behavioral fishing insights
- Predictive fisheries management

The application also creates opportunities for improved communication and data sharing between anglers and marine resource agencies.

Current Limitations and Future Development

Count My Fish is currently a prototype application designed for demonstration and proof-of-concept purposes. The app currently uses SQLite as a local database solution and relies on a limited dataset for predictive modeling. Future improvements include:

- Transitioning to PostgreSQL or cloud-based databases
- Developing a mobile application version
- Integrating official Division of Marine Fisheries datasets
- Adding automated regulation updates
- Expanding predictive modeling capabilities
- Supporting additional fish species
- Tracking invasive species populations

These improvements would help move the application toward a scalable, production-ready platform.

Conclusion

Count My Fish demonstrates how data science, machine learning, and interactive application design can be applied to recreational fishing and fisheries management. By combining real-time data collection, predictive analytics, and community engagement, the app creates value for both anglers and fisheries agencies.

The project highlights the potential for recreational anglers to contribute meaningful environmental and fisheries data while improving their own fishing experience through data-driven insights.